

Exxon Valdez Oil Spill Accident analysis using STAMP and STPA

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Abbreviations

- U.S. - United States
- VTS - Vessel Traffic Service
- VTC - Coast Guard Vessel Tracking Center
- MSO - Marine Safety Office
- UCA – Unsafe Control Action
- AOSC – Alaska Oil Spill Commission
- NTSB – National Transportation Safety Board
- RAYCAS – Raytheon Collision Avoidance System

Introduction

In the evening of March, the 23rd, 1989, an Oil Tanker departed the Alyeska marine terminal on the way to Los Angeles, named the Exxon Valdez. It was owned by the ExxonMobil Shipping Company and transported circa 200984584 liters of crude oil. On this night it was captained by Joseph Hazelwood with the guidance of an Alaskan state pilot and was monitored by the United States (U.S.) Coast Guard with the Vessel Traffic Service (VTS). [1]

Right after midnight, when it was still sailing in Alaska, it suddenly struck the Bligh Reef in Prince William Sound. The vessel being 300 meters long and capable of transporting 1.48 million barrels of crude oil led to this disaster being the worst oil spill from an oil tanker ever at the time. [1]

The accident had multiple factors which contributed to the crash, but one of the main points was the fact that the ExxonMobil Shipping Company had failed to provide a rested crew for the Exxon Valdez, which was also a widespread issue in the industry at the time. Captain Hazelwood has been confirmed to have been sleeping at the time of the accident which led to the third mate maneuvering the vessel to the grounding. Another blaring issue was the malfunctioning Radar system onboard. [1]

This report will analyze the accident reports of the Exxon Valdez and determine xxx!!! Additionally this will be a report with the accident in focus instead of the cleanup, as most other articles on the topic are.

Accident Description

The Oil Tanker Exxon Valdez departed the marine terminal in Valdez, Alaska on the 23rd of March at 9.12 p.m. with the destination of Los Angeles. [1] The route from Valdez terminal out to the sea was one that had been used daily for the past 12 years by over 9000 tankers without a single oil spill. The route went from the bay to the Valdez Arm of Prince William Sound before getting to the sea. This route was controlled by the Coast Guard Vessel Tracking Center (VTC) and the Marine Safety Office (MSO) to regulate the traffic. [2]

At departure the bridge was manned by the captain, the helmsman and the State harbor pilot, but shortly after the third mate relieved the chief mate and additionally, the captain left the bridge. This was, however, against regulations, and the master did not return before the pilot requested him to do so. Closer to midnight the captain responded to an ice report request by the VTC and reported that they would take the inbound channel instead of the outgoing one. In the next half an hour the captain changes the course and the speed several times to navigate through the lane but leaves the third mate to turn the ship back to the traffic lanes after he leaves for “paperwork”. The third mate, who should also have been relieved by the second mate, is not certified to control the ship in coastal operations. The third mate orders several turns to the right but too late, and the ship runs aground at 14 knots. [2], [3]

The ship was operating at a draft of 56 feet while the low tide had caused the reef to be at 30 feet, and the damage to the ship was caused by it hitting the rocky reef at a high speed. The damage surveys done to the ship showed that eight of the eleven cargo tanks on board were torn open during the crash. This led to a loss of 10.1 million gallons of crude oil in just 5 hours, but due to the way that the vessel had grounded, 80% of the cargo stayed on board. [1]

Additionally, the Captain was a known alcoholic, and the Exxon Shipping company was aware of this, and at several points during the trip that led to the accident did other staff note that the captain was drinking. The shipping company, however, decided against acting upon the reports received during the trip. [2]

Control structure

There were seven controllers for the ship in total, four were primary controllers and four were external. The four primary ones were captain Hazelwood, the third mate and second mate that rested each other so count as one, the helmsman and the state pilot. The four external ones were the coast guard services VTS, VTC and MSO, the Exxon shipping management and the maintenance organization. The captain was the main controller of the ship physically while the third mate was responsible for the safety and security on the ship and the second mate was responsible for the ship's navigation. The state pilot was there to guide the captain in piloting the ship and due to regulations, he was required to be there. Finally, the helmsman executed the captains' commands for steering the ship. The chief mate would be a part of the controllers but since he was relieved of duty during the accident he is not regarded in the schematics. [4]

The coast guard services were responsible for monitoring the sea traffic by setting their courses on certain lanes and deciding the speeds. The Exxon shipping management were responsible for manning the ship and having policies for things on the ship like fatigue. The maintenance organization was responsible for the equipment working as intended, such as the radar.[1], [5] The structure can be seen in figure 1 although somewhat simplified.

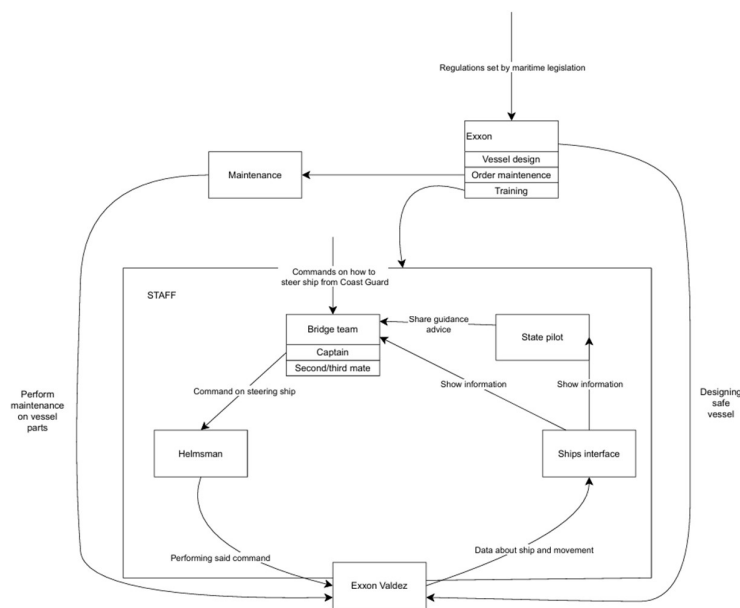


Figure 1: Control Structure of the Exxon Valdez

Explanation of RAYCAS malfunction

Onboard the Exxon Valdez there was a Raytheon Collision Avoidance System (RAYCAS) which was a radar for avoiding underwater collisions. This system was, however, malfunctional on the Exxon Valdez during the night of the collision, but not only that, it had been in this state for a year prior to the incident [1]. The RAYCAS V Radar was not even turned on that fateful night and the whole reason for it not working for over a year was that the Exxon Shipping Company found it too expensive to fix and maintain, which means that they were very aware of it not working. This was, however, not the only RAYCAS radar on board, but it was the more sophisticated one. The other one was working and even used by the third mate just before the collision, which also means that the incident in part could not be purely blamed on the malfunctioning radar, since the third mate had the other required navigation tools and knew where the ship was at all times, even though the most advanced radar was not available. [2], [6]

Hazards and Constraints

Hazard	Constraints
H1: Ship leaves its designated lane of traffic which may lead to a collision or grounding	C1: The ship must stay in its defined lane with a clearance from possible hazards. C2: In cases where deviation from a lane is required a safe re-entry plan must be established.
H2: Unqualified or overworked staff steering the ship may lead to incorrect or delayed maneuvering	C3: The staff at the helm must be qualified to steer the ship in said conditions. C4: Staff must be controlled in a manner so that the bridge and watch teams are both fit for work.
H3: Equipment being unavailable or broken leading to a reduced ability to detect hazards.	C5: Equipment such as navigation machinery must be operational and checked in before departure from a marine terminal. C6: In case of malfunctioning equipment, it must be replaced in other ways by using alternative equipment and reduced speeds.
H4: The VTC not detecting and correcting unsafe movements and intervening	C7: Traffic control must maintain tracking of vessels and possible deviations from planned courses and issue warnings in case of risks.
H5: Too high speeds in tight canals leading to reduced time to correct mistakes and increasing the possible damage in case of a collision	C8: The speed of the vessel must be managed so that there exists a safe margin for corrections in steering.

Table 1: Identified hazards and constraints

Unsafe Control Actions (UCAs)

The unsafe control actions that led to the collision at blight reef were chosen based on those that were found to be the reason according to the final report by the Alaska Oil Spill Commission (AOSC) and the final report by the National Transportation Safety Board (NTSB) [3]. The contributing causes to select the unsafe control actions are:

1. The bridge staff were not at an appropriate level of fitness for the task at hand due to Exxon Shipping company not providing the adequate crew nor schedules leading to overworked staff.
2. The Exxon Shipping company allowed operation with a known defective radar without maintaining its functionality.
3. The captain left the third mate to steer the vessel even though he was not qualified to do so in the conditions present.
4. Course correction commands were issued too late to re-enter the safe lane.
5. Third mate did not wake up the second mate to take over to avoid exhaustion.
6. The VTC did not provide a timely warning about the danger that the vessel was in when it stayed in the wrong lane too long.

UCA analysis table

Control action	Not providing causes hazard	Providing causes hazard	Wrong timing/order causes hazard	Stopped too soon/Applied too long
Enough crew provided to man ship	Crew in danger of being overworked H2	Crew is not qualified to guide the vessel H2	N/A	N/A
Maintaining equipment	Equipment is needed to avoid accident H3	Calibration is done wrong or cheaply which leads to false readings H3	Faulty equipment not fixed before leaving port H3	Vessel and equipment not fully maintenance H3

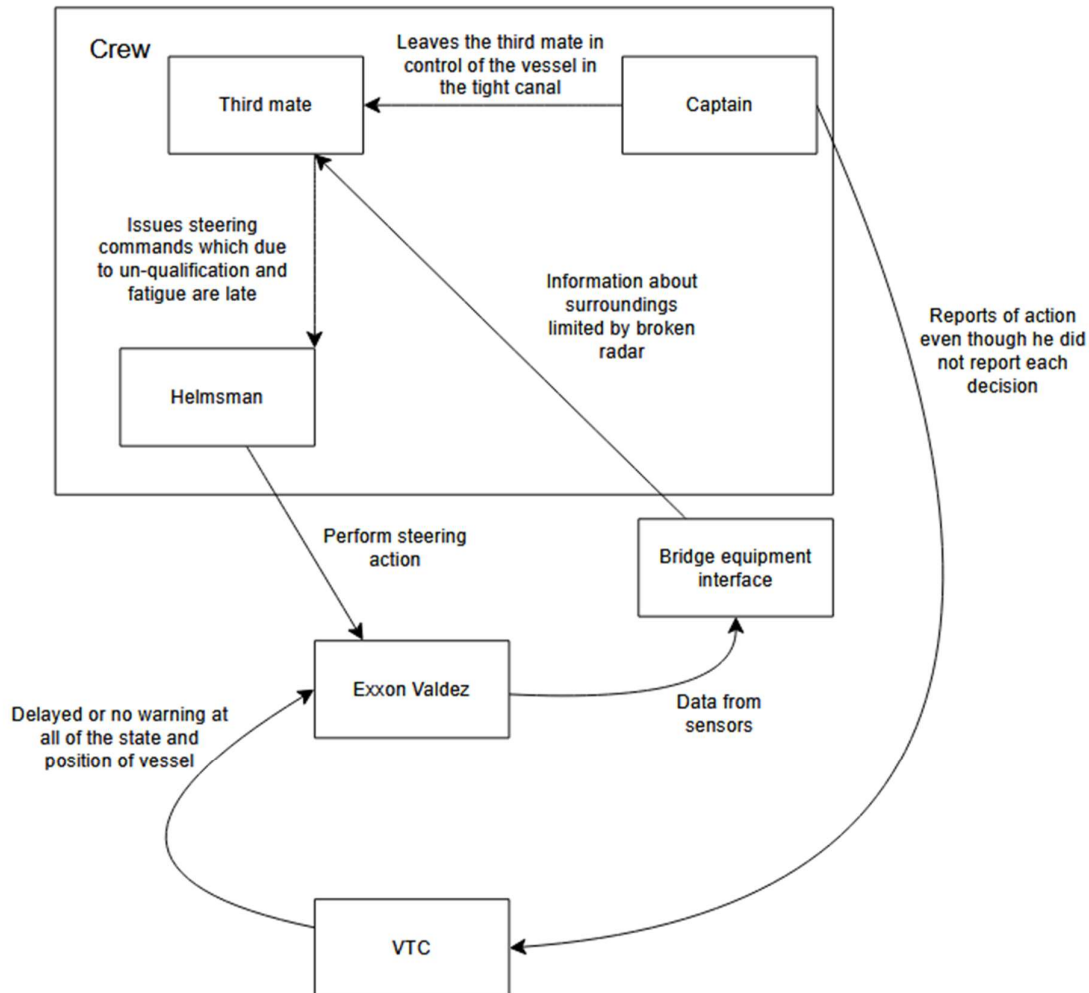
Qualified staff with supervision	Staff make wrong call H1, H2	N/A	Staff make calls too late H1, H5	Staff make unsupervised call H1, H5
Late commands	Giving the wrong commands	Ship remains outside of traffic lane H1	N/A	N/A
Third mate continued working	Poorly handled and bridge left unmanned H1, H5	Staff make bad calls due to being overworked H2	Staff work too long shifts H2	N/A
VTC did not issue warning	Warning is issued incorrectly H4	Vessel and staff don't know of the impending danger H1, H5	Warning issued after point of no return H1	Monitoring would have been needed longer H4

Table 2: Determining UCAs

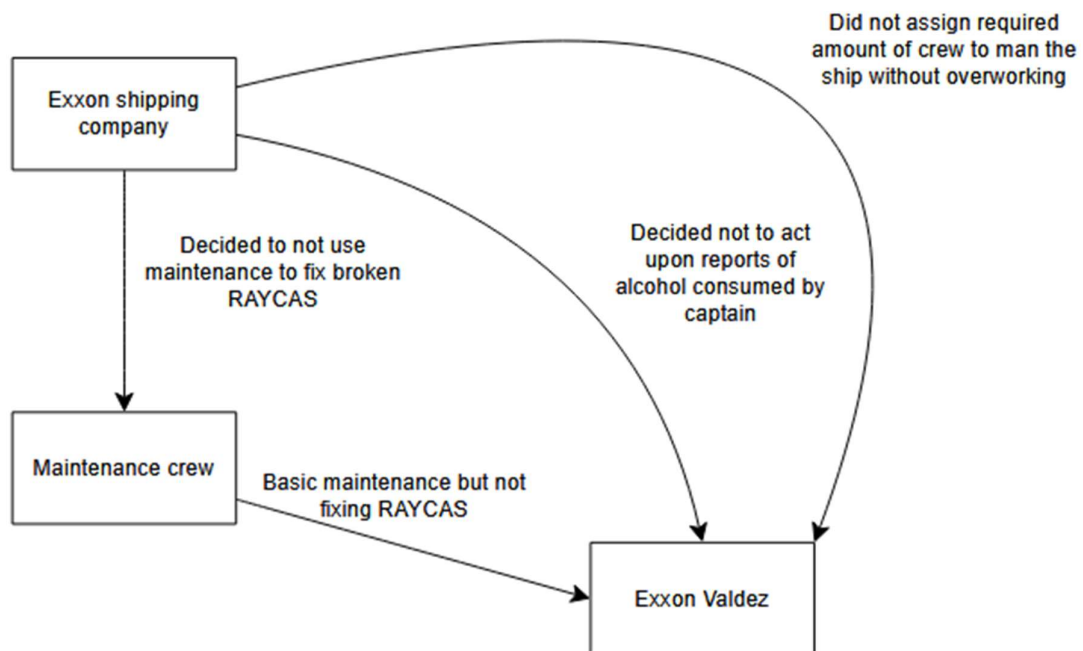
Based on table 2 the identified UCAs are as follows:

1. Fatigue management was not provided adequately, resulting in an overworked bridge team with reduced performance.
2. Critical navigation equipment was not maintained, reducing the crew's ability to detect hazards.
3. Qualified supervision was not ensured when operating in conditions requiring high attention, increasing the likelihood of incorrect or delayed maneuvering.
4. Course correction to re-enter the traffic lane was issued too late leaving insufficient margin to avoid grounding.
5. Relief was not called in time, allowing fatigue to continue into a critical navigation phase.
6. Traffic monitoring did not provide timely warning when the vessel remained outside the safe lane, allowing the unsafe state to persist.

Identifying causes of UCAs



The captain's decision not to accurately tell VTC about decisions made, led to them not being able to warn the vessel about the danger they were in. This decision by the captain was likely to have been made due to him being drunk, as well as leaving the bridge to go sleep. The VTC thus had incorrect information about the location of the Exxon Valdez. The overworked and unqualified third mate that was left in control was too late with the steering commands and was also limited by the broken equipment. [2]



The Exxon shipping company did not supply the appropriate number of crew members for the vessel so that it could work without having to rely on overworked staff. It also did not enact on the report from another Exxon seaman of the captain drinking alcohol during working hours. The maintenance crew did not fix the broken radar since the Exxon shipping company deemed it too expensive. [1]

Defining safety controls

The table below is one of new safety controls based on the constraints

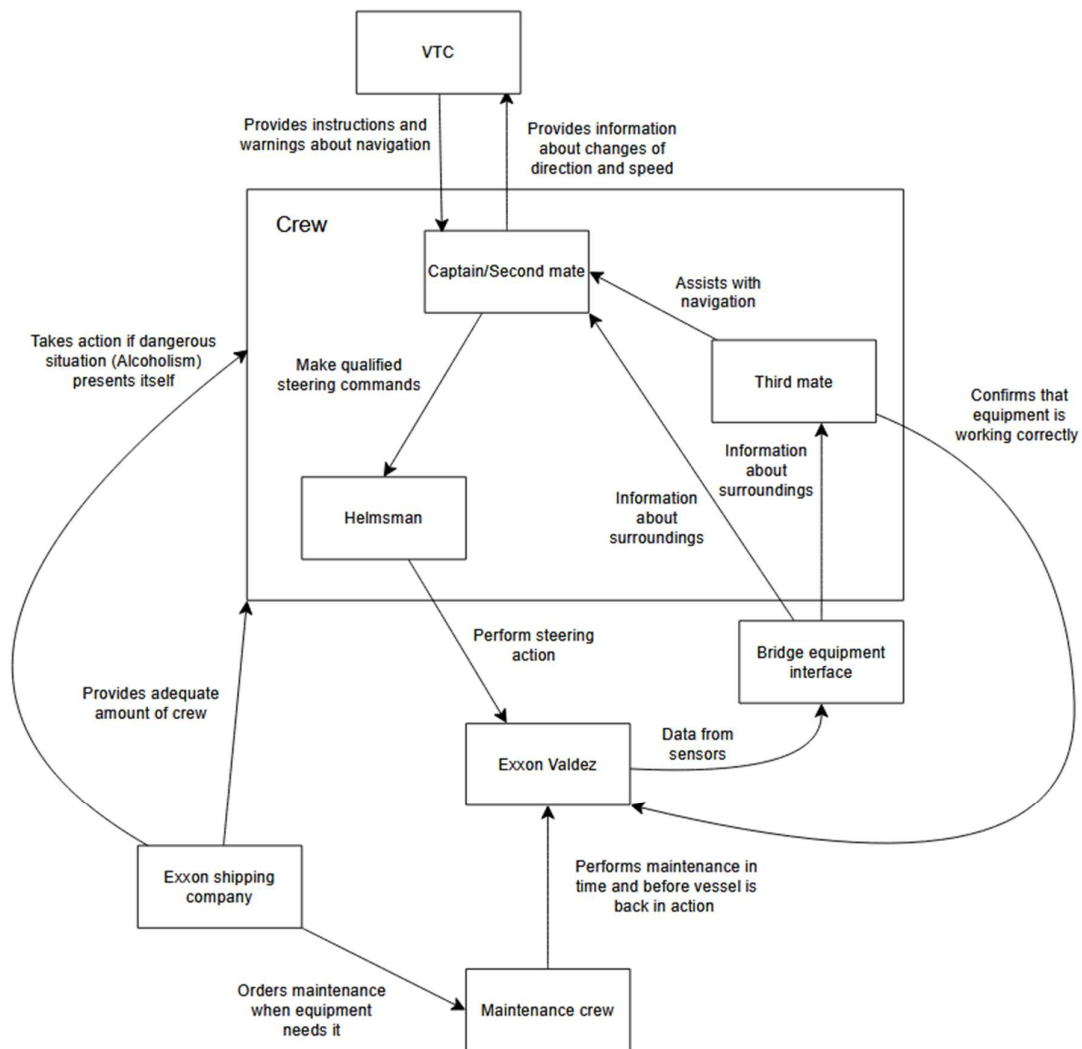


Figure 4: A safer control structure

This structure is done in such a way that future similar accidents could be avoided. This is done by mitigating the multiple factors that led to the accident such as not allowing overworked crew to navigate the ship and by not letting an unqualified staff member steer the vessel. The recommendations are as follows:

1. The Exxon Shipping company needs to act if a report is made about a possible danger, such as a crew member being intoxicated.

2. The Exxon Shipping company must provide an adequate amount of crew members to avoid overworking.
3. The Exxon Shipping company must make sure that all equipment is serviced on time.
4. The captain must not leave unqualified staff to steer the vessel
5. The captain must inform the VTC about changes in speed and steering
6. The third mate shall assist with navigation, not do it by themselves

References

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